Andhra University College of Engineering

Computer Science and Engineering | Course Code: CS301

Operating Systems - Laboratory Manual & Practical Guide

Syllabus Focus: CS301 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Process Management & CPU Scheduling

" Syllabus Scope: Process states, Process Control Block PCB, Context switching overhead, Scheduling algor

Unit 2: Inter-Process Communication & Synchronization

" Syllabus Scope: Critical section problem, Peterson solution, Hardware test-and-set, Counting and binary

CS301: Operating Systems - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Deadlocks & Mitigation Strategies

" Deadlock conditions, Resource allocation graphs, Bankers algorithm for safety and avoidance, Detection algor

Unit 4: Memory Management & Demand Paging

" Logical vs physical address space, Contiguous allocation, Paging, Inverted page tables, Segmentation, Demand

Unit 5: File Systems & Storage Subsystems

" File concepts, Directory structures, Inode allocation methods contiguous, linked, indexed, Free-space manage

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Implementation of Process Creation and IPC via POSIX Pipes and Shared Memory in C

Lab 2: Simulation of CPU Scheduling Algorithms FCFS, SJF, Priority, and Round Robin

Lab 3: Thread Synchronization using POSIX Mutexes and Semaphores for Producer-Consumer problem

Lab 4: Simulation of Banker Algorithm for Deadlock Avoidance and Page Replacement FIFO and LRU

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Abraham Silberschatz, Peter B. Galvin - Operating System Concepts Wiley 10th Ed
* Andrew S. Tanenbaum - Modern Operating Systems Pearson 4th Ed
* William Stallings - Operating Systems: Internals and Design Principles Pearson

Authorized NoteVault Educational Document " Verified for Registered University Students